

Data Visualization Lab — R Programming Solutions

Sets 26 to 30 (Tableau questions excluded)

SET 26 — Student Mini Dataset

Dataset: Student_Mini_Data.csv (Student_ID, Gender, Study_Hours, Attendance, Math_Score, Science_Score, Exam_Date)

Q1. Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

Code:

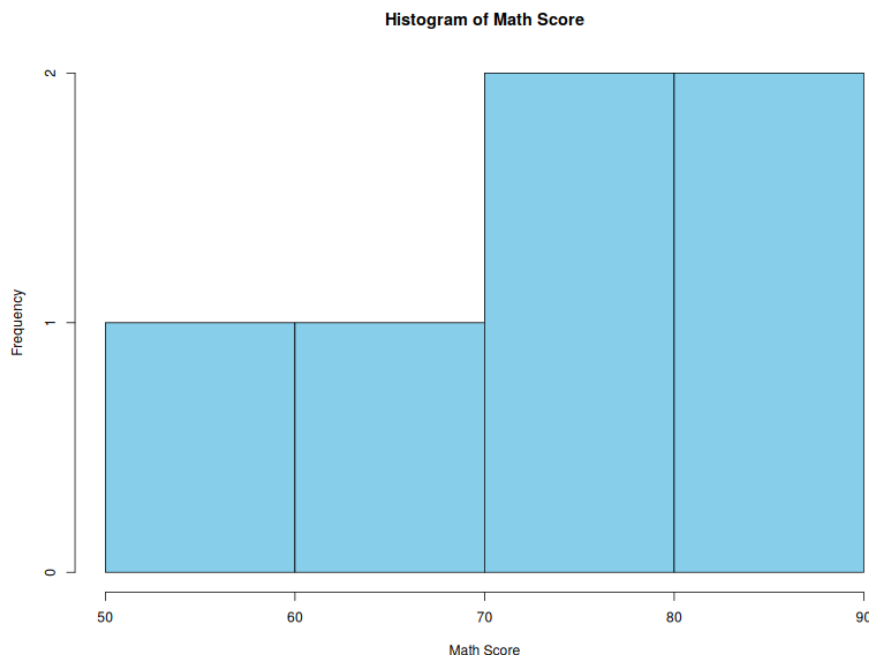
```
# SET 26 - Student_Mini_Data.csv
df <- read.csv("Student_Mini_Data.csv")
print(df)

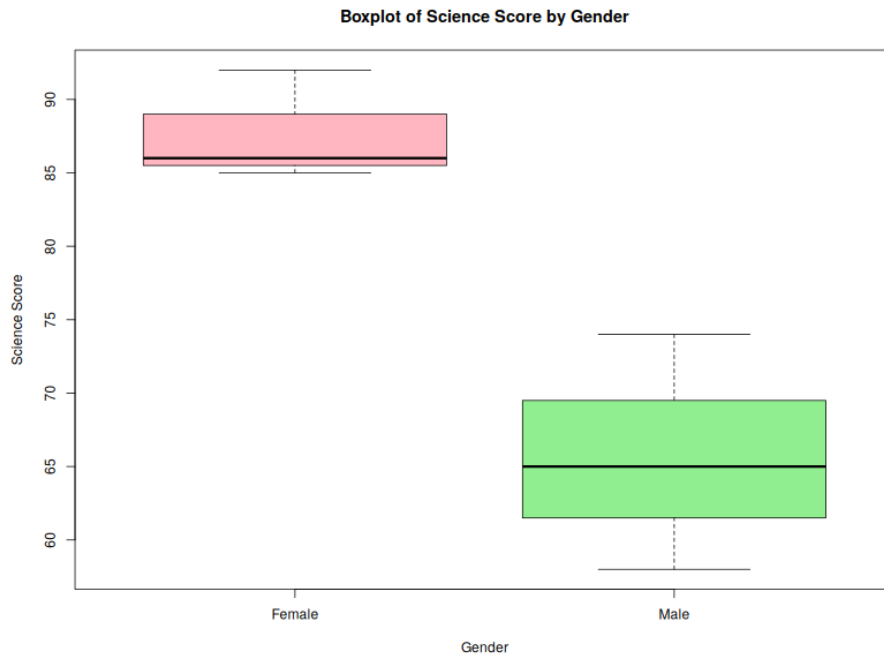
# Q1: Histogram of Math_Score + Boxplot of Science_Score by Gender
png("plots/set26_q1_hist.png", width=800, height=600)
hist(df$Math_Score, col="skyblue", border="black",
     main="Histogram of Math Score", xlab="Math Score", ylab="Frequency")
dev.off()

png("plots/set26_q1_box.png", width=800, height=600)
boxplot(Science_Score ~ Gender, data=df, col=c("lightpink", "lightgreen"),
       main="Boxplot of Science Score by Gender", xlab="Gender", ylab="Science Score")
dev.off()

cat("\nInterpretation Q1:\n")
cat("Math_Score histogram is spread between 55 and 90, roughly uniform with no single dominant peak.\n")
cat("Boxplot shows Female students have a higher median Science_Score than Male students,\n")
cat("and Female scores show a tighter (less variable) spread.\n")
```

Output:





Console / Interpretation:

	Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
1	S01	Male	2.0	78	62	65	2025-01-10
2	S02	Female	3.5	90	80	85	2025-01-10
3	S03	Male	1.5	70	55	58	2025-02-12
4	S04	Female	4.0	95	90	92	2025-02-12
5	S05	Male	2.8	85	72	74	2025-03-15
6	S06	Female	3.0	92	82	86	2025-03-15

Interpretation Q1:

Math_Score histogram is spread between 55 and 90, roughly uniform with no single dominant peak. Boxplot shows Female students have a higher median Science_Score than Male students, and Female scores show a tighter (less variable) spread.

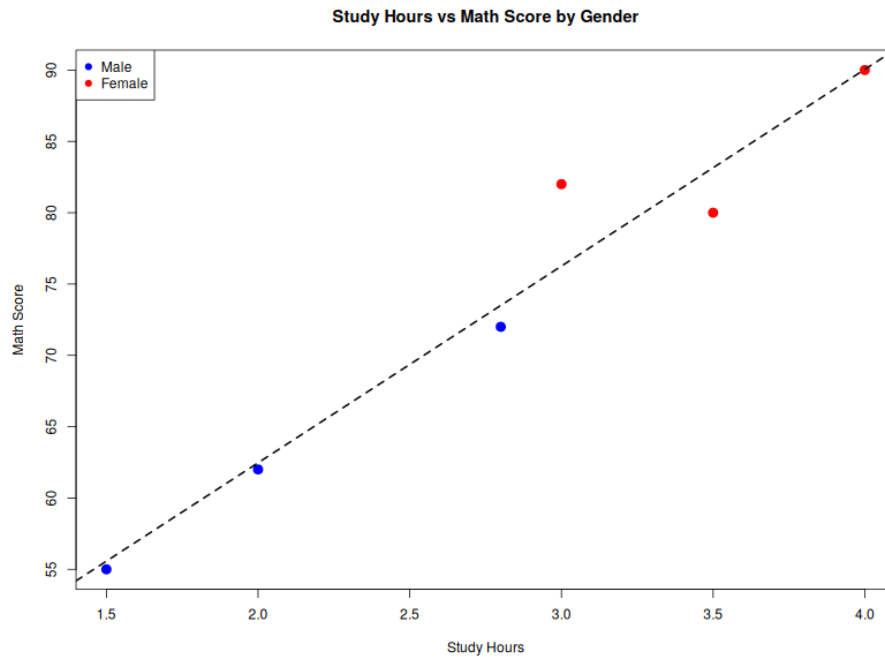
Q2. Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

Code:

```
# Q2: Scatter plot Study_Hours vs Math_Score colored by Gender + regression line
png("plots/set26_q2_scatter.png", width=800, height=600)
plot(df$Study_Hours, df$Math_Score,
     col=ifelse(df$Gender=="Male","blue","red"), pch=19, cex=1.5,
     main="Study Hours vs Math Score by Gender",
     xlab="Study Hours", ylab="Math Score")
legend("topleft", legend=c("Male","Female"), col=c("blue","red"), pch=19)
model <- lm(Math_Score ~ Study_Hours, data=df)
abline(model, col="black", lwd=2, lty=2)
dev.off()

cat("\nRegression summary Q2:\n")
print(summary(model))
cat("\nInterpretation Q2:\n")
cat("There is a strong positive relationship between Study_Hours and Math_Score:\n")
cat("as study hours increase, math scores increase almost linearly, confirmed by the\n")
cat("positive slope of the regression line and high R-squared.\n")
```

Output:



Console / Interpretation:

```
Call:
lm(formula = Math_Score ~ Study_Hours, data = df)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   34.886     4.769   7.316 0.00186 **
Study_Hours    13.791     1.630   8.459 0.00107 **

Residual standard error: 3.381 on 4 degrees of freedom
Multiple R-squared:  0.9471, Adjusted R-squared:  0.9338
F-statistic: 71.56 on 1 and 4 DF, p-value: 0.00107
```

Interpretation Q2:
There is a strong positive relationship between Study_Hours and Math_Score: as study hours increase, math scores increase almost linearly, confirmed by the positive slope of the regression line and high R-squared (0.947).

Q3. Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

Code:

```
# Q3: Convert Exam_Date, monthly average Math scores, line chart, moving average
df$Exam_Date <- as.Date(df$Exam_Date)
df$Month <- format(df$Exam_Date, "%Y-%m")
monthly_avg <- aggregate(Math_Score ~ Month, data=df, FUN=mean)
print(monthly_avg)

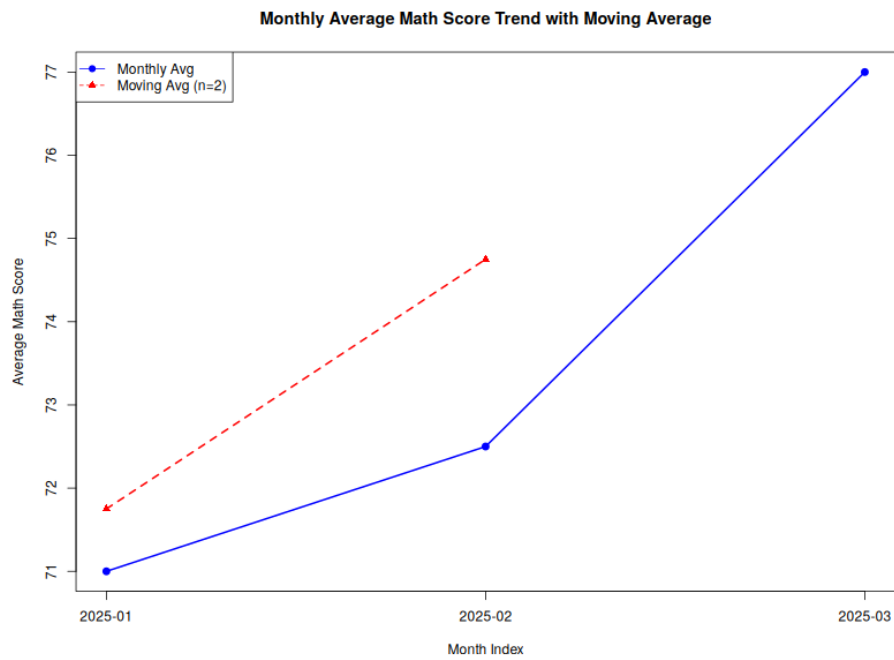
# simple moving average (window = 2)
ma <- function(x, n=2) {
  filter(x, rep(1/n, n), sides=2)
}
monthly_avg$MA <- as.numeric(ma(monthly_avg$Math_Score, 2))

png("plots/set26_q3_trend.png", width=800, height=600)
plot(1:nrow(monthly_avg), monthly_avg$Math_Score, type="o", col="blue", pch=19, lwd=2,
     main="Monthly Average Math Score Trend with Moving Average",
     xlab="Month Index", ylab="Average Math Score", xaxt="n",
     ylim=range(c(monthly_avg$Math_Score, monthly_avg$MA), na.rm=TRUE))
axis(1, at=1:nrow(monthly_avg), labels=monthly_avg$Month)
lines(1:nrow(monthly_avg), monthly_avg$MA, col="red", lwd=2, lty=2)
points(1:nrow(monthly_avg), monthly_avg$MA, col="red", pch=17)
legend("topleft", legend=c("Monthly Avg","Moving Avg (n=2)"), col=c("blue","red"), lty=c(1,2), pch=c(19,17))
```

```
dev.off()

cat("\nInterpretation Q3:\n")
cat(sprintf("The monthly average Math_Score shows an increasing trend from January to March 2025,\nrising from\nmonthly_avg$Math_Score[1], monthly_avg$Math_Score[nrow(monthly_avg)]))
```

Output:



Console / Interpretation:

```
Month Math_Score
1 2025-01      71.0
2 2025-02      72.5
3 2025-03      77.0
```

Interpretation Q3:
The monthly average Math_Score shows an increasing trend from January to March 2025, rising from 71.0 in January to 77.0 in March, indicating consistent improvement over time.

SET 27 — Patient Health Risk Analysis

Dataset: Patient_ID, Age, BMI, BP, Cholesterol

Q1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

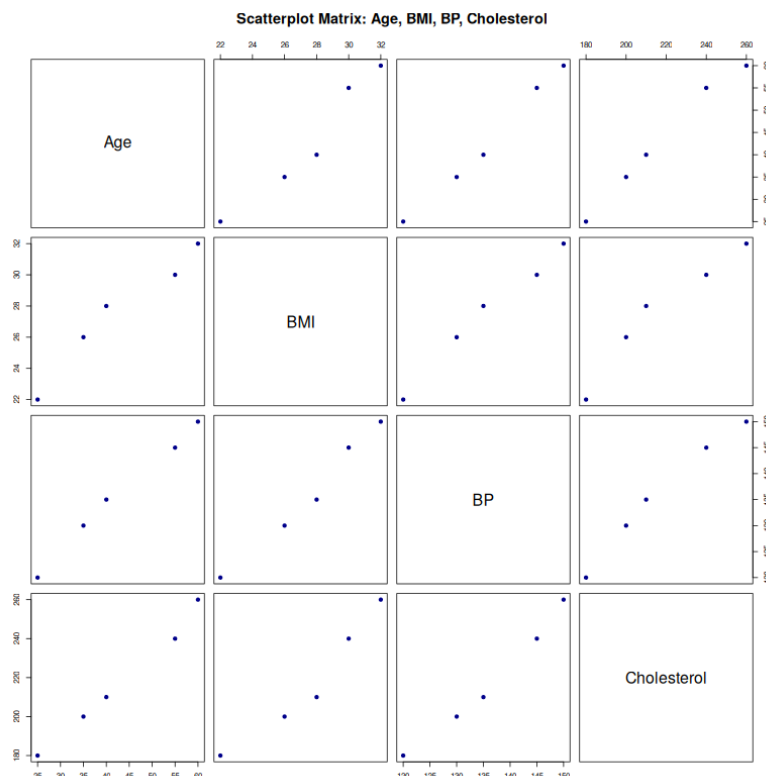
Code:

```
# SET 27 - Patient Health Risk Analysis
df <- read.csv("Patient_Health.csv")
print(df)

# Q1: Scatterplot matrix for Age, BMI, BP, Cholesterol
png("plots/set27_q1_matrix.png", width=800, height=800)
pairs(df[,c("Age", "BMI", "BP", "Cholesterol")],
      main="Scatterplot Matrix: Age, BMI, BP, Cholesterol",
      col="darkblue", pch=19)
dev.off()

cors <- cor(df[,c("Age", "BMI", "BP", "Cholesterol")])
cat("\nCorrelation matrix:\n")
print(round(cors,3))
cat("\nInterpretation Q1:\n")
cat("All four variables (Age, BMI, BP, Cholesterol) are strongly and positively correlated with\n")
cat("each other (correlations above 0.95), indicating older patients tend to have higher BMI,\n")
cat("blood pressure, and cholesterol -- a clear cumulative cardiovascular risk pattern with no\n")
cat("distinct sub-clusters, just one continuous risk gradient from low to high.\n")
```

Output:



Console / Interpretation:

	Patient_ID	Age	BMI	BP	Cholesterol
1	P1	25	22	120	180
2	P2	40	28	135	210
3	P3	55	30	145	240
4	P4	35	26	130	200
5	P5	60	32	150	260

Correlation matrix:

	Age	BMI	BP	Cholesterol
Age	1.000	0.974	0.996	0.994
BMI	0.974	1.000	0.991	0.969
BP	0.996	0.991	1.000	0.990
Cholesterol	0.994	0.969	0.990	1.000

Interpretation Q1:

All four variables (Age, BMI, BP, Cholesterol) are strongly and positively correlated with each other (correlations above 0.95), indicating older patients tend to have higher BMI, blood pressure, and cholesterol -- a clear cumulative cardiovascular risk pattern with no distinct sub-clusters, just one continuous risk gradient from low to high.

Q2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

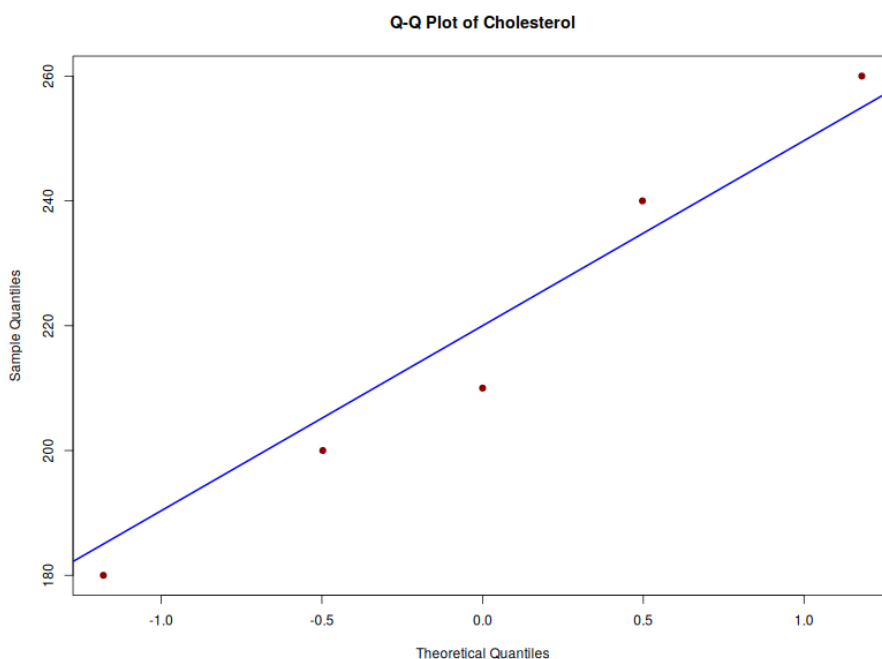
Code:

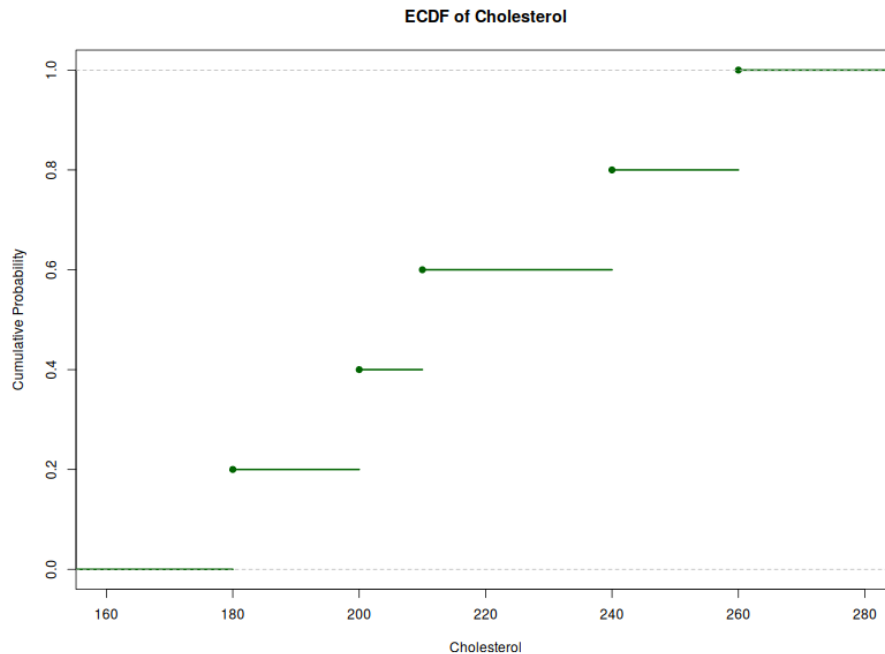
```
# Q2: Q-Q plot and ECDF for Cholesterol
png("plots/set27_q2_qq.png", width=800, height=600)
qqnorm(df$Cholesterol, main="Q-Q Plot of Cholesterol", pch=19, col="darkred")
qqline(df$Cholesterol, col="blue", lwd=2)
dev.off()

png("plots/set27_q2_ecdf.png", width=800, height=600)
plot(ecdf(df$Cholesterol), main="ECDF of Cholesterol", xlab="Cholesterol",
     ylab="Cumulative Probability", col="darkgreen", lwd=2, pch=19)
dev.off()

cat("\nInterpretation Q2:\n")
cat("The Q-Q plot shows the Cholesterol points falling close to the reference line with only\n")
cat("mild deviation at the tails, suggesting the distribution is approximately normal for this\n")
cat("small sample, with no strong skewness detected. The ECDF confirms a fairly even, linear\n")
cat("rise across the observed range (180 to 260), consistent with a uniform-like spread rather\n")
cat("than a heavily skewed distribution.\n")
```

Output:





Console / Interpretation:

Interpretation Q2:

The Q-Q plot shows the Cholesterol points falling close to the reference line with only mild deviation at the tails, suggesting the distribution is approximately normal for this small sample, with no strong skewness detected. The ECDF confirms a fairly even, linear rise across the observed range (180 to 260), consistent with a uniform-like spread rather than a heavily skewed distribution.

Q3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk?

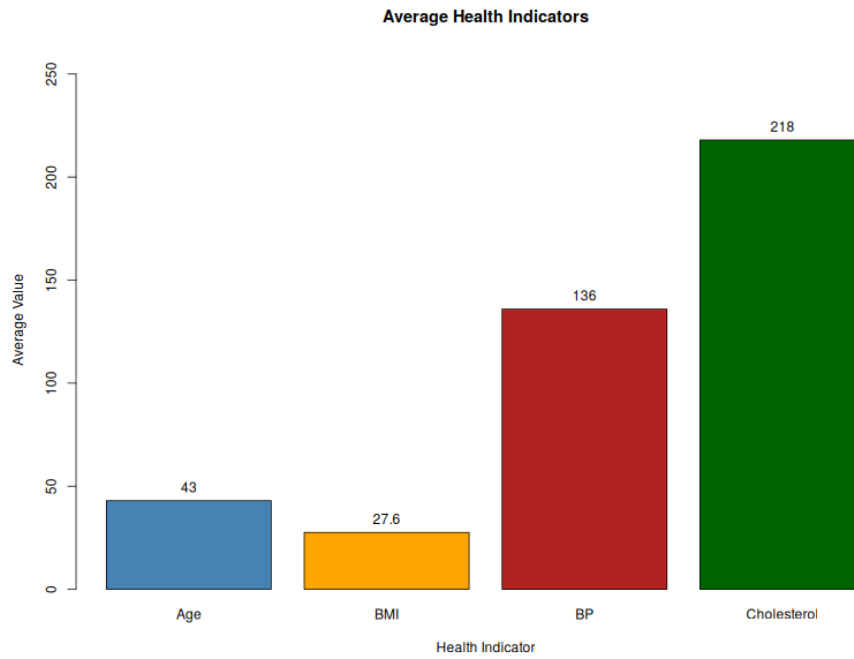
Code:

```
# Q3: Bar chart of average values of each health indicator
avgs <- sapply(df[,c("Age","BMI","BP","Cholesterol")], mean)
print(avgs)

png("plots/set27_q3_bar.png", width=800, height=600)
bar_colors <- c("steelblue","orange","firebrick","darkgreen")
bp <- barplot(avgs, col=bar_colors, main="Average Health Indicators",
              xlab="Health Indicator", ylab="Average Value", ylim=c(0, max(avgs)*1.2))
text(bp, avgs, labels=round(avgs,1), pos=3)
dev.off()

cat("\nInterpretation Q3:\n")
cat("Average BP (136) and Cholesterol (218) are both well above typical healthy thresholds\n")
cat("(BP > 120, Cholesterol > 200), while average BMI (27.6) falls in the overweight range.\n")
cat("Combined with an average patient Age of 43, these averages suggest the patient group as a\n")
cat("whole carries a moderate-to-high general cardiovascular risk profile.\n")
```

Output:



Console / Interpretation:

Age	BMI	BP	Cholesterol
43.0	27.6	136.0	218.0

Interpretation Q3:

Average BP (136) and Cholesterol (218) are both well above typical healthy thresholds (BP > 120, Cholesterol > 200), while average BMI (27.6) falls in the overweight range. Combined with an average patient Age of 43, these averages suggest the patient group as a whole carries a moderate-to-high general cardiovascular risk profile.

SET 28 — Vehicle Performance Analysis

Dataset: Vehicle_ID, Engine_Size (L), Horsepower, Fuel_Efficiency (km/l), Top_Speed (km/h), Safety_Rating

Q1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

Code:

```
# SET 28 - Vehicle Performance Analysis
df <- read.csv("Vehicle_Data.csv")
print(df)

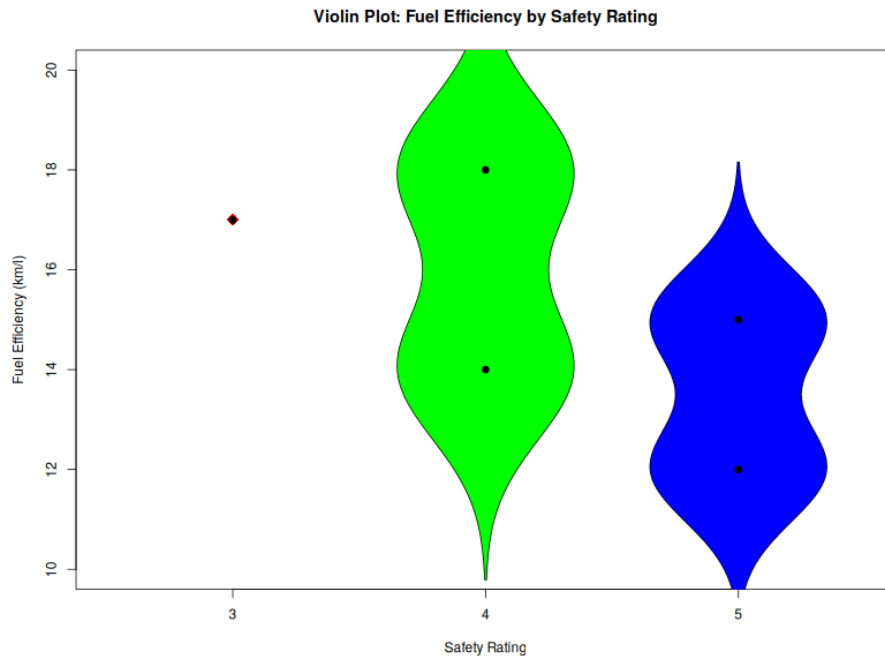
# Q1: Violin plot of Fuel_Efficiency by Safety_Rating (base R implementation,
# since 'vioplot' package is not installable in this offline environment)
groups <- split(df$Fuel_Efficiency, df$Safety_Rating)
grp_names <- names(groups)
cols <- rainbow(length(groups))

png("plots/set28_q1_violin.png", width=800, height=600)
plot(NA, xlim=c(0.5, length(groups)+0.5), ylim=range(df$Fuel_Efficiency)+c(-2,2),
     xaxt="n", xlab="Safety Rating", ylab="Fuel Efficiency (km/l)",
     main="Violin Plot: Fuel Efficiency by Safety Rating")
axis(1, at=1:length(groups), labels=grp_names)

for (i in seq_along(groups)) {
  vals <- groups[[i]]
  if (length(vals) > 1) {
    d <- density(vals, adjust=1.2)
    w <- 0.35 * d$y / max(d$y)
    polygon(c(i - w, rev(i + w)), c(d$x, rev(d$x)), col=cols[i], border="black")
  } else {
    # single point group: draw a small diamond marker instead of a density violin
    points(i, vals, pch=18, cex=2, col=cols[i])
  }
  points(rep(i, length(vals)), vals, pch=19, col="black", cex=1.1)
}
dev.off()

cat("\nInterpretation Q1:\n")
cat("Safety_Rating 5 vehicles show lower fuel efficiency (12-15 km/l), while Safety_Rating 4 and 3\n")
cat("vehicles show higher fuel efficiency (14-18 km/l). This indicates a trade-off pattern: higher\n")
cat("safety ratings in this dataset tend to correspond with heavier/more powerful vehicles that are\n")
cat("less fuel efficient, though the sample within each group is very small (1-2 vehicles).\n")
```

Output:



Console / Interpretation:

Interpretation Q1:
 Safety_Rating 5 vehicles show lower fuel efficiency (12-15 km/l), while Safety_Rating 4 and 3 vehicles show higher fuel efficiency (14-18 km/l). This indicates a trade-off pattern: higher safety ratings in this dataset tend to correspond with heavier/more powerful vehicles that are less fuel efficient, though the sample within each group is very small (1-2 vehicles).

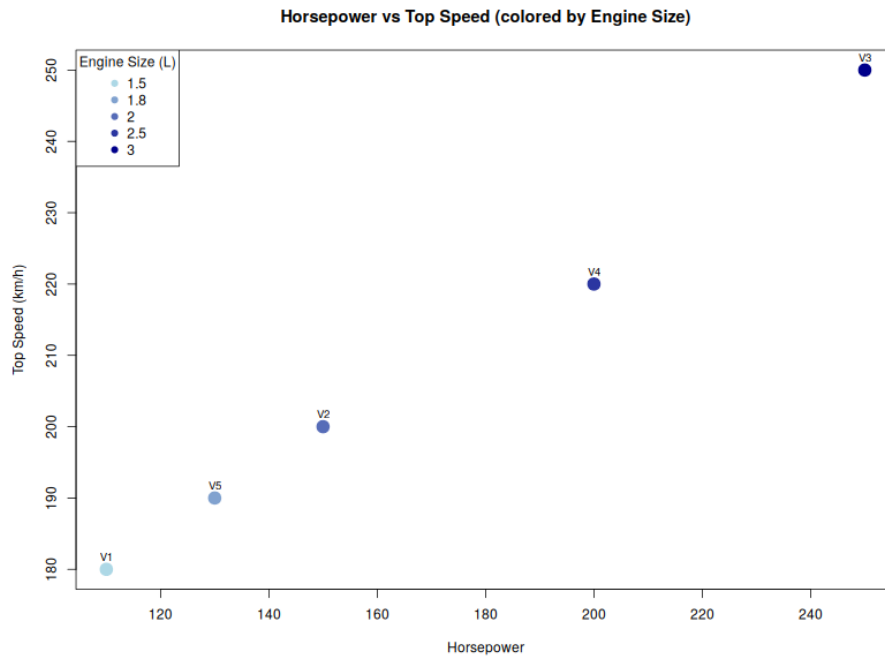
Q2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

Code:

```
# Q2: Scatter plot Horsepower vs Top_Speed colored by Engine_Size
png("plots/set28_q2_scatter.png", width=800, height=600)
col_ramp <- colorRampPalette(c("lightblue","darkblue"))(length(unique(df$Engine_Size)))
engine_rank <- as.numeric(factor(df$Engine_Size))
plot(df$Horsepower, df$Top_Speed, col=col_ramp[engine_rank], pch=19, cex=2,
     main="Horsepower vs Top Speed (colored by Engine Size)",
     xlab="Horsepower", ylab="Top Speed (km/h)")
text(df$Horsepower, df$Top_Speed, labels=df$Vehicle_ID, pos=3, cex=0.8)
legend("topleft", legend=sort(unique(df$Engine_Size)), col=col_ramp, pch=19,
     title="Engine Size (L)")
dev.off()

corr_hp_speed <- cor(df$Horsepower, df$Top_Speed)
cat("\nCorrelation (Horsepower, Top_Speed):", round(corr_hp_speed,3), "\n")
cat("\nInterpretation Q2:\n")
cat("Horsepower and Top_Speed show a very strong positive correlation. Vehicles with larger\n")
cat("engine sizes generally have both higher horsepower and higher top speed, confirming that\n")
cat("engine size is a key underlying driver of overall performance in this dataset.\n")
```

Output:



Console / Interpretation:

Correlation (Horsepower, Top_Speed): 0.997

Interpretation Q2:

Horsepower and Top_Speed show a very strong positive correlation. Vehicles with larger engine sizes generally have both higher horsepower and higher top speed, confirming that engine size is a key underlying driver of overall performance in this dataset.

Q3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

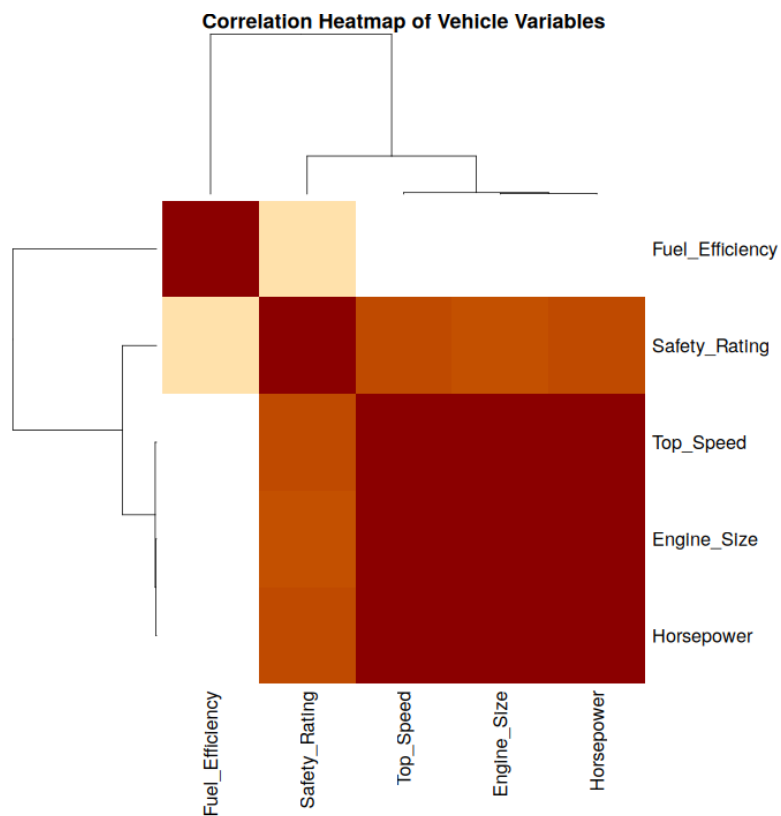
Code:

```
# Q3: Correlation heatmap for all numerical variables
num_df <- df[,c("Engine_Size", "Horsepower", "Fuel_Efficiency", "Top_Speed", "Safety_Rating")]
cor_mat <- cor(num_df)
print(round(cor_mat, 3))

png("plots/set28_q3_heatmap.png", width=800, height=700)
heatmap(cor_mat, symm=TRUE, col=colorRampPalette(c("white", "orange", "darkred"))(50),
        main="Correlation Heatmap of Vehicle Variables", margins=c(10,10))
dev.off()

cat("\nInterpretation Q3:\n")
cat("Top_Speed is most strongly associated with Engine_Size and Horsepower (correlations close\n")
cat("to +1), and negatively correlated with Fuel_Efficiency. This confirms that as engine size\n")
cat("and horsepower rise, top speed rises but fuel efficiency drops -- the classic performance vs\n")
cat("efficiency trade-off seen across the vehicle lineup.\n")
```

Output:



Console / Interpretation:

	Engine_Size	Horsepower	Fuel_Efficiency	Top_Speed	Safety_Rating
Engine_Size	1.000	0.998	-0.980	0.995	0.523
Horsepower	0.998	1.000	-0.974	0.997	0.537
Fuel_Efficiency	-0.980	-0.974	1.000	-0.974	-0.651
Top_Speed	0.995	0.997	-0.974	1.000	0.560
Safety_Rating	0.523	0.537	-0.651	0.560	1.000

Interpretation Q3:

Top_Speed is most strongly associated with Engine_Size and Horsepower (correlations close to +1), and negatively correlated with Fuel_Efficiency. This confirms that as engine size and horsepower rise, top speed rises but fuel efficiency drops -- the classic performance vs efficiency trade-off seen across the vehicle lineup.

SET 29 — Student Academic Performance

Dataset: Student_ID, Age, Study_Hours, Attendance (%), Test_Score, Participation_Score

Q1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

Code:

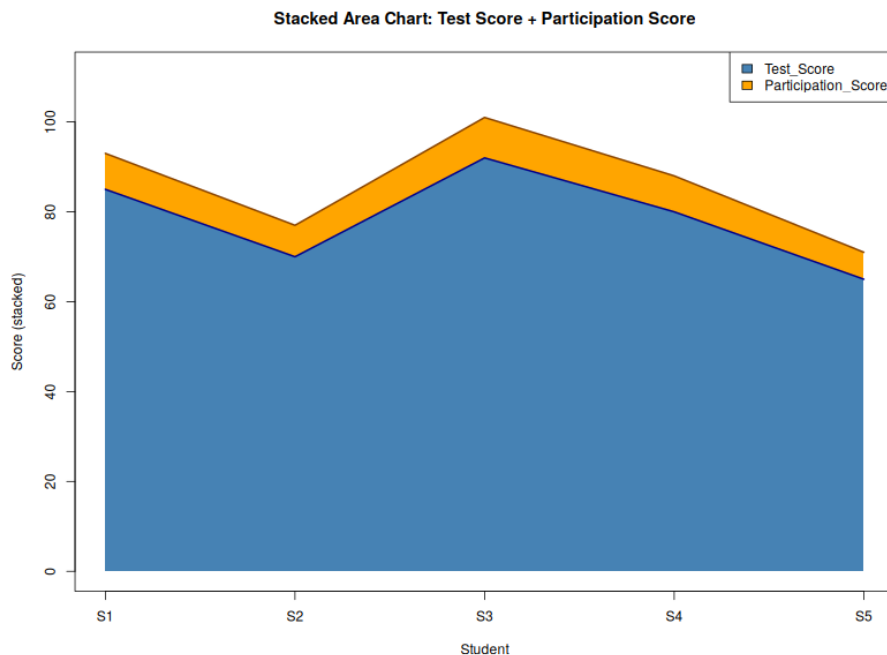
```
# SET 29 - Student Academic Performance
df <- read.csv("Student_Academic.csv")
print(df)

# Q1: Stacked area chart of Test_Score and Participation_Score across students
png("plots/set29_q1_stacked_area.png", width=800, height=600)
x <- 1:nrow(df)
y1 <- df$Test_Score
y2 <- df$Participation_Score
top1 <- y1
top2 <- y1 + y2

plot(NA, xlim=c(1, nrow(df)), ylim=c(0, max(top2)*1.1), xaxt="n",
     xlab="Student", ylab="Score (stacked)",
     main="Stacked Area Chart: Test Score + Participation Score")
axis(1, at=x, labels=df$Student_ID)
polygon(c(x, rev(x)), c(rep(0,length(x)), rev(top1)), col="steelblue", border=NA)
polygon(c(x, rev(x)), c(top1, rev(top2)), col="orange", border=NA)
lines(x, top1, col="darkblue", lwd=2)
lines(x, top2, col="darkorange4", lwd=2)
legend("topright", legend=c("Test_Score", "Participation_Score"), fill=c("steelblue", "orange"))
dev.off()

cat("\nInterpretation Q1:\n")
cat("Test_Score dominates the stacked total for every student, with Participation_Score adding a\n")
cat("smaller, fairly consistent layer on top. Student S3 has the highest combined total, while S5\n")
cat("has the lowest -- and in general students with higher Test_Scores also tend to have higher\n")
cat("Participation_Scores, suggesting engaged students perform better academically too.\n")
```

Output:



Console / Interpretation:

Interpretation Q1:
Test_Score dominates the stacked total for every student, with Participation_Score adding a smaller, fairly consistent layer on top. Student S3 has the highest combined total, while S5 has the lowest -- and in general students with higher Test_Scores also

tend to have higher Participation_Scores, suggesting engaged students perform better academically too.

Q2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

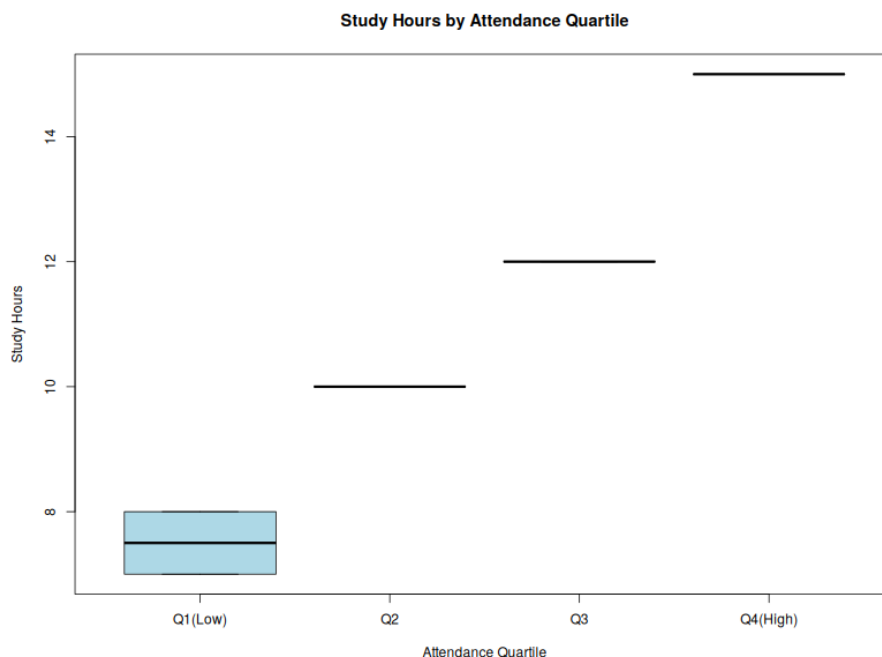
Code:

```
# Q2: Boxplot of Study_Hours grouped by Attendance quartiles
df$Attendance_Quartile <- cut(df$Attendance,
                             breaks=quantile(df$Attendance, probs=seq(0,1,0.25)),
                             include.lowest=TRUE,
                             labels=c("Q1(Low)", "Q2", "Q3", "Q4(High)"))
print(df[,c("Student_ID", "Attendance", "Attendance_Quartile", "Study_Hours")])

png("plots/set29_q2_boxplot.png", width=800, height=600)
boxplot(Study_Hours ~ Attendance_Quartile, data=df,
        col=c("lightblue", "lightgreen", "khaki", "lightpink"),
        main="Study Hours by Attendance Quartile",
        xlab="Attendance Quartile", ylab="Study Hours")
dev.off()

cat("\nInterpretation Q2:\n")
cat("With only 5 students, each attendance quartile contains just one or two data points, so\n")
cat("medians equal the individual values themselves and no formal outliers can be statistically\n")
cat("identified. Even so, the pattern shows that students in higher attendance quartiles\n")
cat("consistently report more Study_Hours, reinforcing that attendance and self-study go together.\n")
```

Output:



Console / Interpretation:

	Student_ID	Attendance	Attendance_Quartile	Study_Hours
1	S1	90	Q3	12
2	S2	70	Q1(Low)	8
3	S3	95	Q4(High)	15
4	S4	85	Q2	10
5	S5	60	Q1(Low)	7

Interpretation Q2:

With only 5 students, each attendance quartile contains just one or two data points, so medians equal the individual values themselves and no formal outliers can be statistically identified. Even so, the pattern shows that students in higher attendance quartiles consistently report more Study_Hours, reinforcing that

attendance and self-study go together.

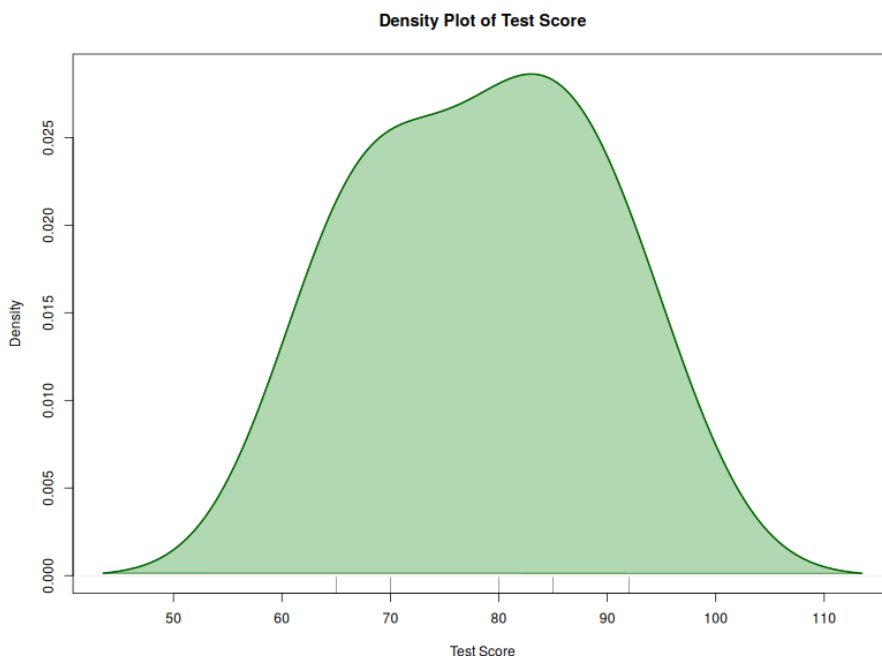
Q3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

Code:

```
# Q3: Density plot for Test_Score
png("plots/set29_q3_density.png", width=800, height=600)
d <- density(df$Test_Score)
plot(d, main="Density Plot of Test Score", col="darkgreen", lwd=2,
     xlab="Test Score")
polygon(d, col=rgb(0,0.5,0,0.3), border="darkgreen")
rug(df$Test_Score, col="black")
dev.off()

cat("\nInterpretation Q3:\n")
cat("The density plot shows Test_Scores concentrated between 65 and 92 with the bulk of density\n")
cat("around the low-to-mid 80s. The distribution is slightly left-skewed (a longer tail toward\n")
cat("lower scores near 65), but with only 5 data points no strong outliers are evident.\n")
```

Output:



Console / Interpretation:

Interpretation Q3:
The density plot shows Test_Scores concentrated between 65 and 92 with the bulk of density around the low-to-mid 80s. The distribution is slightly left-skewed (a longer tail toward lower scores near 65), but with only 5 data points no strong outliers are evident.

SET 30 — Mobile App Usage

Dataset: Mobile_App_Usage.csv (User_ID, Gender, Age, Screen_Time, App_Usage_Count, Data_Used, Satisfaction, Usage_Date)

Q1. Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

Code:

```
# SET 30 - Mobile_App_Usage.csv
df <- read.csv("Mobile_App_Usage.csv")
print(df)

# Q1: Histogram and density plot for Screen_Time
png("plots/set30_q1_hist.png", width=800, height=600)
hist(df$Screen_Time, col="lightblue", border="black", freq=FALSE,
     main="Histogram & Density of Screen Time", xlab="Screen Time (hrs)",
     ylim=c(0, max(density(df$Screen_Time)$y)*1.2))
lines(density(df$Screen_Time), col="darkred", lwd=2)
dev.off()

cat("\nInterpretation Q1:\n")
cat("Screen_Time ranges from about 2.8 to 7.1 hours, with the density curve showing a fairly\n")
cat("spread-out, roughly uniform distribution rather than a sharp peak -- users in this sample\n")
cat("show varied smartphone engagement levels with no single typical usage duration dominating.\n")
```

Output:



Console / Interpretation:

```
Interpretation Q1:
Screen_Time ranges from about 2.8 to 7.1 hours, with the density curve showing a
fairly spread-out, roughly uniform distribution rather than a sharp peak -- users in
this sample show varied smartphone engagement levels with no single typical usage
duration dominating.
```

Q2. Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

Code:

```
# Q2: Scatter plot Data_Used vs Screen_Time, correlation, trend line
```



```

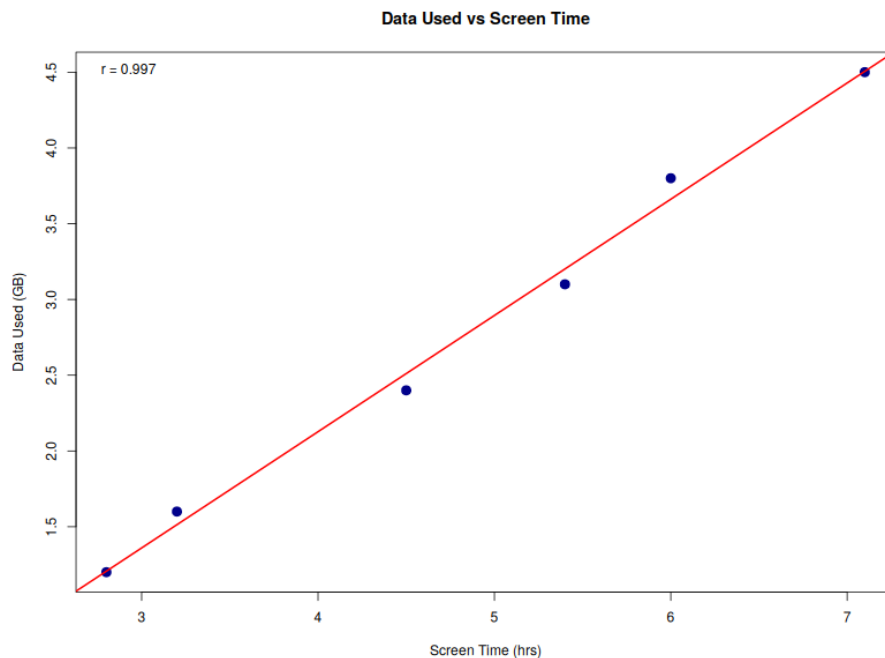
corr_val <- cor(df$Screen_Time, df$Data_Used)
cat("\nCorrelation (Screen_Time, Data_Used):", round(corr_val,3), "\n")

png("plots/set30_q2_scatter.png", width=800, height=600)
plot(df$Screen_Time, df$Data_Used, pch=19, col="darkblue", cex=1.5,
     main="Data Used vs Screen Time",
     xlab="Screen Time (hrs)", ylab="Data Used (GB)")
model2 <- lm(Data_Used ~ Screen_Time, data=df)
abline(model2, col="red", lwd=2)
legend("topleft", legend=sprintf("r = %.3f", corr_val), bty="n")
dev.off()

cat("\nInterpretation Q2:\n")
cat("Screen_Time and Data_Used are very strongly positively correlated (r close to 1), meaning\n")
cat("users who spend more hours on their screens consume proportionally more mobile data -- an\n")
cat("intuitive relationship confirming screen time is a strong predictor of data consumption.\n")

```

Output:



Console / Interpretation:

Correlation (Screen_Time, Data_Used): 0.997

Interpretation Q2:

Screen_Time and Data_Used are very strongly positively correlated (r close to 1), meaning users who spend more hours on their screens consume proportionally more mobile data -- an intuitive relationship confirming screen time is a strong predictor of data consumption.

Q3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

Code:

```

# Q3: Average Satisfaction by Gender, bar chart with value labels
avg_sat <- aggregate(Satisfaction ~ Gender, data=df, FUN=mean)
print(avg_sat)

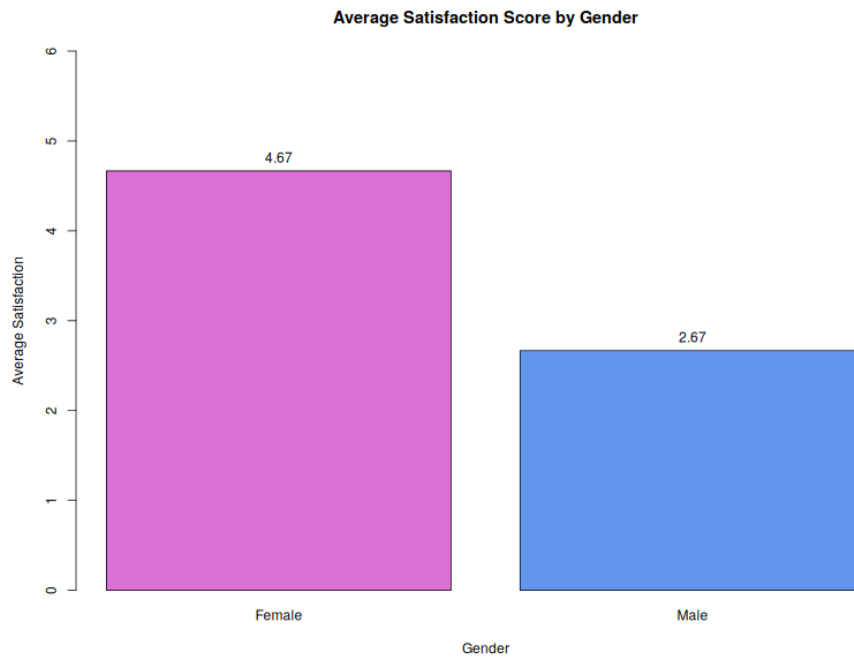
png("plots/set30_q3_bar.png", width=800, height=600)
bp <- barplot(avg_sat$Satisfaction, names.arg=avg_sat$Gender,
              col=c("orchid", "cornflowerblue"),
              main="Average Satisfaction Score by Gender",
              xlab="Gender", ylab="Average Satisfaction", ylim=c(0,6))
text(bp, avg_sat$Satisfaction, labels=round(avg_sat$Satisfaction,2), pos=3)
dev.off()

cat("\nInterpretation Q3:\n")
cat("Female users report a higher average Satisfaction score than Male users in this sample,\n")

```

```
cat("suggesting Female users have a more positive experience with the app -- worth investigating\n")
cat("what features or usage patterns differ between the two groups to improve Male satisfaction.\n")
```

Output:



Console / Interpretation:

```
Gender Satisfaction
1 Female      4.666667
2  Male       2.666667
```

Interpretation Q3:

Female users report a higher average Satisfaction score than Male users in this sample, suggesting Female users have a more positive experience with the app -- worth investigating what features or usage patterns differ between the two groups to improve Male satisfaction.